0111 — The kernel floor is documented, not renegotiated

- **Status:** Accepted — documentation implemented
- **Date:** 2026-09-03
- **Milestone / issue:** #620 — the sandbox requires Linux 6.12 (Landlock ABI 6), which no current mainstream LTS ships, and it was documented nowhere.
- **Supersedes:** nothing. **Reaffirms** 0029's premise rather than reopening it — see "What this ADR does not do," below.
- **Related:** 0029 (INV-13: no degraded tier when Strict is unavailable — the decision this ADR's cost falls out of); the git-version-floor treatment in `docs/SUPPORTED_VERSIONS.md` (the sibling precedent this ADR follows for *documenting* a floor, though the enforcement mechanism differs — see below).

Context

`crates/git-vista-server/src/sandbox/mod.rs:175` declares the sandbox's Landlock floor:

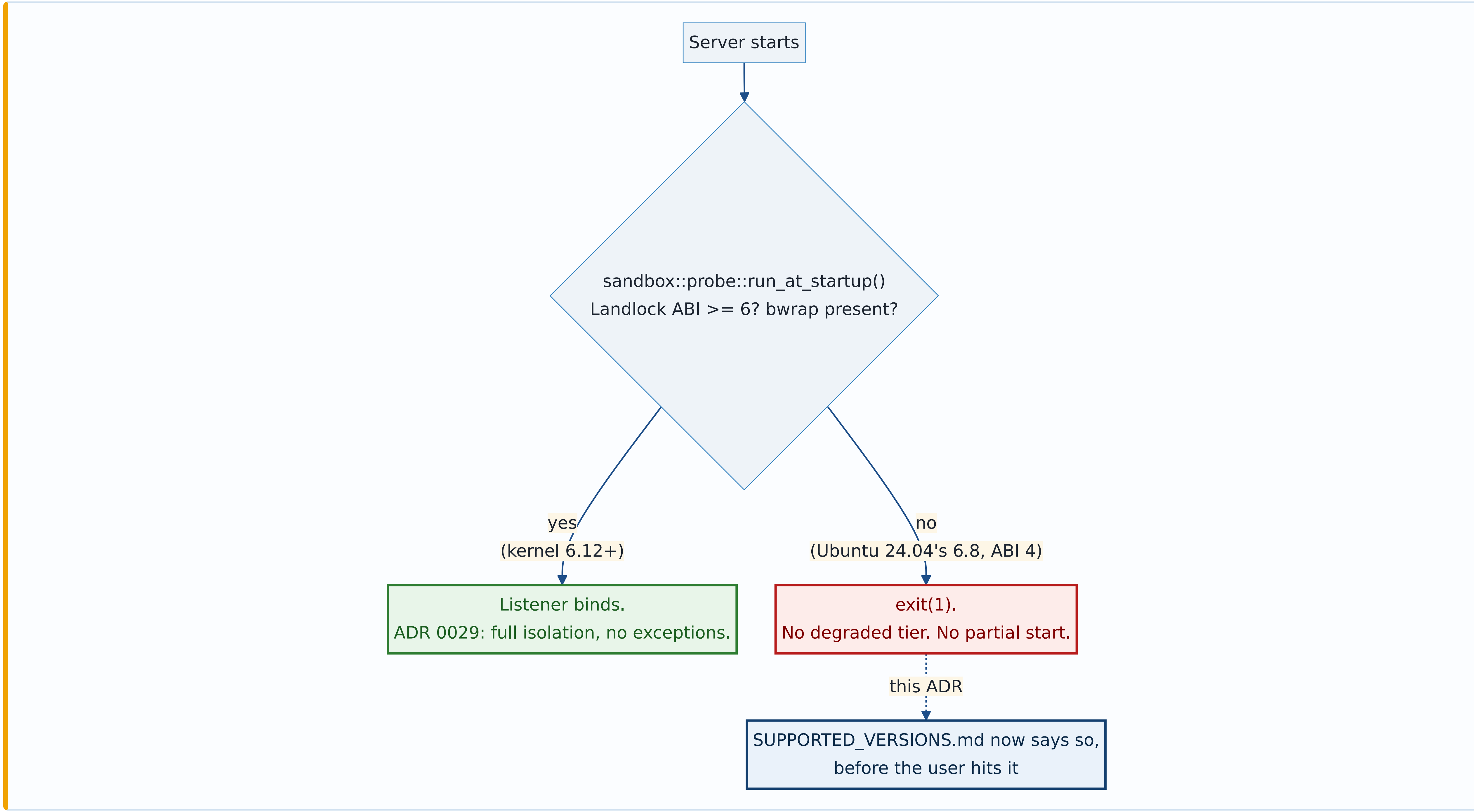
```
pub(crate) const LANDLOCK_ABI_FLOOR: u32 = 6;
```

`main.rs:218` gates every server start on it, with no degraded path, deliberately, per ADR 0029:

```
if let Err(refusal) = sandbox::probe::run_at_startup().await {
    eprintln!("error: {refusal}");
    std::process::exit(1);
}
```

Landlock ABI 6 first ships in **Linux 6.12** (November 2024). `#620` measured a stock Ubuntu 24.04 cloud image on titan directly — `syscall(444, NULL, 0, 1)` — and got `landlock_abi=4` on kernel `6.8.0-138-generic`. So on the current Ubuntu LTS, supported by Canonical until 2029, `Capabilities::strict_missing()` returns `["landlock_abi>=6", "bwrap"]`, `run_at_startup()` returns `Err(CapabilityAbsent)`, and the server exits 1 before executing a byte of repository content.

Nothing was wrong with the code. `docs/SUPPORTED_VERSIONS.md` documented a git floor and a Safari floor, derived the same way this ADR derives the kernel one, and said nothing about the kernel at all. A user on the most common LTS in the world would hit a boot refusal with no way to have known it was coming.



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The ladder, and what ships it

From the kernel's own Landlock ABI history, plus each distribution's shipped kernel (only the Ubuntu 24.04 row is directly measured; the rest are derived from public release notes):

Landlock ABI	First kernel
2	5.19
3	6.2
4	6.7
5	6.10
6 — this floor	**6.12**

Distribution	GA kernel	ABI	Starts?
Ubuntu 22.04 LTS	5.15	1	No
Debian 12 bookworm	6.1	2	No
RHEL 9	5.14	1	No
Ubuntu 24.04 LTS	6.8	4 (**measured**)	No
Debian 13 trixie	6.12	6	Yes, exactly at the floor
Ubuntu 26.04	7.0	8	Yes

Every current mainstream LTS except Debian 13 fails the probe. This is not a narrow edge case — it is the common path.

The decision, exactly as taken

The kernel floor is a documented requirement, not a bug, and ADR 0029's premise is not reopened. A hard-fail sandbox with no degraded tier is only honest if the host it hard-fails on was foreseeable to the person installing it. Today it wasn't; that was the actual defect, and it is a documentation gap, not an argument for a fallback path.

Tom's call (this ADR): **documented requirement**, not a bug to fix by relaxing INV-13. The remedy for a user on Ubuntu 24.04 is an HWE kernel (`linux-generic-hwe-24.04`, which is 6.14 on 24.04.3 — clears the floor) plus `apt install bubblewrap`, and that remedy is now written down.

`docs/SUPPORTED_VERSIONS.md` gains a `## Linux kernel: 6.12 or later` section alongside the existing git and Safari floors, carrying the ladder and distribution tables above, the measured Ubuntu 24.04 result, and the HWE remedy.

Why this is not enforced the same way the git floor is

`docs/SUPPORTED_VERSIONS.md`'s git floor is a *documented number a CI job parses and rebuilds* (ADR 0082): the heading is the one source of truth, and a check fails the build if it drifts from what the code enforces. The kernel floor does not get that same treatment, and that is a difference in mechanism, not in rigor:

- The git floor is a version comparison against an installed binary that CI can build fresh cheaply (`make` from source, a few minutes).
- The kernel floor is a property of the machine the *server itself* runs on — there is no "install a different kernel and test against it" step CI can cheaply take, and #620 confirms this cannot

even be simulated: a container shares the host kernel, so `ubuntu:22.04` under rootless podman on titan reports the *host's* Landlock ABI (8), not the guest's. No container can ever test this floor.

- `LANDLOCK_ABI_FLOOR` already lives in exactly one place (`sandbox/mod.rs:175`), and `run_at_startup()` already enforces it, live, on every boot, on the real running kernel — which is a stronger guarantee than a CI-time comparison against a number in a doc heading. The gap #620 found was never "the floor isn't enforced." It was "the floor isn't *documented*." This ADR and its doc change close that gap; they do not add a second enforcement mechanism, because the existing one is already the more direct check.

What this ADR does not do

- **It does not lower the floor.** ABI 6 stays 6; the reasoning for it (`sandbox/mod.rs:172-174`, "the signal and abstract-unix-socket scopes this design uses") is unchanged and unexamined here.
- **It does not add a degraded mode.** ADR 0029's INV-13 — hard-fail, no exceptions — is reaffirmed, not revisited. A boot refusal on an unsupported kernel is *working as designed*; documenting that it will happen is the entire scope of this decision.
- **It does not claim HWE-kernel or Debian-13 clearance was independently re-measured here.** Both are derived from the ABI ladder and each distribution's own published kernel version, consistent with how #620 scoped its own "not yet measured" section. Only the Ubuntu 24.04 stock result is a direct measurement.

Consequences

- A user reading `docs/SUPPORTED_VERSIONS.md` before installing on Ubuntu 24.04 now sees the boot refusal coming and the exact remedy, instead of meeting `error: CapabilityAbsent` with no context.
- The refusal message itself (`sandbox::probe`'s output) already names what is absent; this ADR and the doc change are about the moment *before* install, which nothing previously addressed.
- No code changes. No test changes. No new enforcement surface — the existing boot-time probe remains the single source of truth, live rather than a doc-parsed comparison.